



1. DDL (Data Definition Language)

✓ CREATE DATABASE

```
CREATE DATABASE aslab;  
USE aslab;
```

✓ CREATE TABLE (PK, AI)

```
CREATE TABLE jurusan (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    nama_jurusan VARCHAR(50),  
    gedung VARCHAR(10)  
);
```

✓ FOREIGN KEY

```
CREATE TABLE mahasiswa (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    nim VARCHAR(20) UNIQUE,  
    nama VARCHAR(50),  
    kelas VARCHAR(10),  
    jurusan_id INT,  
    FOREIGN KEY (jurusan_id) REFERENCES jurusan(id)  
);
```



2. DML (Insert, Update, Delete)

✓ INSERT MULTI-ROW

```
INSERT INTO jurusan (nama_jurusan, gedung) VALUES  
( 'Teknik Informatika', 'G1'),  
( 'Sistem Informasi', 'G2'),  
( 'Teknologi Informasi', 'G3');
```

✓ UPDATE

```
UPDATE mahasiswa  
SET kelas = 'IF-47-02'  
WHERE id = 1;
```

✓ DELETE

```
DELETE FROM mahasiswa  
WHERE nim = '103032300101';
```



3. SELECT

✓ SELECT semua kolom

```
SELECT * FROM mahasiswa;
```

✓ SELECT dengan filter

```
SELECT * FROM mahasiswa  
WHERE kelas = 'IF-47-02';
```



4. JOIN

✓ JOIN 3 Tabel

```
SELECT m.nama, mk.nama_matkul, n.nilai  
FROM nilai n  
JOIN mahasiswa m ON m.id = n.mahasiswa_id  
JOIN matkul mk ON mk.id = n.matkul_id;
```

✓ JOIN + WHERE

```
SELECT m.nama, mk.nama_matkul, n.nilai  
FROM nilai n  
JOIN mahasiswa m ON m.id = n.mahasiswa_id  
JOIN matkul mk ON mk.id = n.matkul_id  
WHERE n.nilai < 80;
```



5. GROUP BY

✓ Rata-rata nilai per jurusan

```
SELECT j.nama_jurusan, AVG(n.nilai) AS rata  
FROM nilai n  
JOIN mahasiswa m ON m.id = n.mahasiswa_id  
JOIN jurusan j ON j.id = m.jurusan_id  
GROUP BY j.id;
```

✓ Rata-rata per mata kuliah

```
SELECT mk.nama_matkul, AVG(n.nilai)  
FROM nilai n  
JOIN matkul mk ON mk.id = n.matkul_id
```

GROUP BY mk.id;



6. HAVING (Filter setelah GROUP BY)

✓ Mata kuliah dengan rata-rata > 80

```
SELECT mk.nama_matkul, AVG(n.nilai) AS rata
FROM nilai n
JOIN matkul mk ON mk.id = n.matkul_id
GROUP BY mk.id
HAVING AVG(n.nilai) > 80;
```

✓ Mahasiswa yang ambil lebih dari 1 matkul

```
SELECT m.nama, COUNT(*) AS jumlah_matkul
FROM nilai n
JOIN mahasiswa m ON m.id = n.mahasiswa_id
GROUP BY m.id
HAVING COUNT(*) > 1;
```



7. WHERE vs HAVING — Cheat Sheet Singkat

Keterangan	WHERE	HAVING
Filter baris	✓	✗
Filter hasil GROUP	✗	✓
Boleh pakai AVG/COUNT	✗	✓
Eksekusi	sebelum GROUP	setelah GROUP



8. SUBQUERY

✓ Subquery hasil 1 nilai (global average)

```
SELECT m.nama, n.nilai
FROM nilai n
JOIN mahasiswa m ON m.id = n.mahasiswa_id
WHERE n.nilai > (
    SELECT AVG(nilai)
    FROM nilai
);
```

9. CORRELATED SUBQUERY (Per Matkul) — Level Sulit

✓ Nilai di atas rata-rata matkul masing-masing

```
SELECT m.nama, mk.nama_matkul, n.nilai
FROM nilai n
JOIN mahasiswa m ON m.id = n.mahasiswa_id
JOIN matkul mk ON mk.id = n.matkul_id
WHERE n.nilai > (
    SELECT AVG(n2.nilai)
    FROM nilai n2
    WHERE n2.matkul_id = n.matkul_id
);
```

📌 **Intinya:**
Subquery punya akses ke baris luar → n.matkul_id.

10. CTE (Common Table Expression)

✓ CTE dasar

```
WITH gabungan AS (
    SELECT m.nama, mk.nama_matkul, n.nilai
    FROM nilai n
    JOIN mahasiswa m ON m.id = n.mahasiswa_id
    JOIN matkul mk ON mk.id = n.matkul_id
)
SELECT *
FROM gabungan
WHERE nilai > 85;
```

✓ CTE + GROUP BY

```
WITH rata_nilai AS (
    SELECT mahasiswa_id, AVG(nilai) AS rata
    FROM nilai
    GROUP BY mahasiswa_id
)
SELECT m.nama, r.rata
FROM rata_nilai r
JOIN mahasiswa m ON m.id = r.mahasiswa_id;
```

11. TEMPLATE SUPER RINGKAS (Bisa Hafalin)

✓ JOIN 3 tabel

```
SELECT ...  
FROM nilai n  
JOIN mahasiswa m ON m.id = n.mahasiswa_id  
JOIN matkul mk ON mk.id = n.matkul_id;
```

✓ GROUP BY

```
SELECT kolom, AGG()  
FROM tabel  
GROUP BY kolom;
```

✓ HAVING

```
HAVING AGG() > value;
```

✓ SUBQUERY

```
WHERE kolom > (SELECT ...)
```

✓ CTE

```
WITH nama_cte AS (  
    SELECT ...  
)  
SELECT * FROM nama_cte;
```

CHEAT SHEET ALTER TABLE (FULL VERSION)

1. Menambah Kolom

✓ Tambah kolom baru

```
ALTER TABLE mahasiswa  
ADD email VARCHAR(50);
```

✓ Tambah kolom di posisi tertentu

```
ALTER TABLE mahasiswa  
ADD alamat VARCHAR(100) AFTER nama;
```

2. Mengubah Tipe Kolom

✓ Modify tipe data

```
ALTER TABLE mahasiswa  
MODIFY nama VARCHAR(100);
```

✓ Ubah kolom menjadi NOT NULL

```
ALTER TABLE mahasiswa  
MODIFY kelas VARCHAR(10) NOT NULL;
```

3. Mengubah Nama Kolom

```
ALTER TABLE mahasiswa  
CHANGE nama nama_lengkap VARCHAR(50);
```

4. Menghapus Kolom

```
ALTER TABLE mahasiswa  
DROP COLUMN alamat;
```

5. Menambah FOREIGN KEY

```
ALTER TABLE matkul  
ADD CONSTRAINT fk_jurusan_matkul  
FOREIGN KEY (jurusan_id)  
REFERENCES jurusan(id);
```

6. Menghapus FOREIGN KEY

Pertama cek nama FK:

```
SHOW CREATE TABLE matkul;
```

Lalu hapus:

```
ALTER TABLE matkul  
DROP FOREIGN KEY fk_jurusan_matkul;
```

7. Mengubah Nama Tabel

```
RENAME TABLE mahasiswa TO data_mahasiswa;
```

8. Menghapus Constraint UNIQUE

Pertama cek namanya:

```
SHOW CREATE TABLE mahasiswa;
```

Biasanya muncul seperti:

```
UNIQUE KEY nim (nim)
```

Untuk hapus:

```
ALTER TABLE mahasiswa  
DROP INDEX nim;
```

● 9. Menambah Kolom dengan Default Value

```
ALTER TABLE mahasiswa  
ADD status VARCHAR(10) DEFAULT 'Aktif';
```

● 10. Mengubah Default Value

```
ALTER TABLE mahasiswa  
ALTER status SET DEFAULT 'Nonaktif';
```

● 11. Menghapus Default Value

```
ALTER TABLE mahasiswa  
ALTER status DROP DEFAULT;
```

● 12. Set kolom Autoincrement

```
ALTER TABLE nilai  
MODIFY id INT AUTO_INCREMENT;
```

● 13. Menghapus Primary Key

```
ALTER TABLE nilai  
DROP PRIMARY KEY;
```

● 14. Menambah Primary Key

```
ALTER TABLE nilai  
ADD PRIMARY KEY (id);
```